

MURAD MEGJHANI, PhD

Associate Research Scientist, Dept. of Neurology • Columbia University Irving Medical Center
Roman Family Faculty, Dept. of Computer Science • Barnard College, Columbia University
710 W 168th St, New York, NY 10032 • mm5025@cumc.columbia.edu • 408.310.6264
<https://megjhani.github.io/>

EDUCATION

Columbia University Medical Center, New York, NY, United States Postdoc, Department of Neurology Advisors: Dr. Soojin Park Topic: Real-Time Detection of Delayed Cerebral Ischemia	2021
University of Houston, Houston TX, United States PhD, Electrical & Computer Engineering Advisors: Dr. Badrinath Roysam, Dr. Jose-Luis Contreras Vidal Dissertation: Data-Driven, Label Consistent, Dictionary Learning Methods for Analysis of Biological Datasets.	2016
Chaitanya Bharthi Institute of Technology, India B.E., Electrical & Computer Engineering.	2005

RESEARCH INTERESTS

Machine learning for physiological time series • non-invasive intracranial pressure estimation • Neurocritical care AI • Foundation models for clinical data • Federated learning • Calibration-aware prediction • EEG-based brain-state modeling • Safe AI for high-stakes clinical workflows • Causal inference • Computational neuroscience

GRANTS & FUNDING

In Review

	Funder	Title	Role	Total Budget	My Budget	Dates
G13	American Heart Association (Innovative Project Award)	Neural Resonance Mapping for State-Dependent Acoustic Modulation in Migraine	Principal Investigator	300,000	300,000	Submitted

Active

	Funder	Title	Role	Total Budget	My Budget	Dates
G12	American Heart Association (Second Century Faculty Independence Award)	Non-Invasive ICP Waveform Derivation Using Machine Learning	Principal Investigator	300,000	300,000	2024–2027

G11	National Institutes of Health (NIH R01 NS131606)	Machine Learning to Optimize Management of Acute Hydrocephalus	Co-Investigator	3,248,657	180,000	2023–2028
G10	National Institutes of Health (NIH R01 NS129760)	COSMIC: ContinuOuS Monitoring Tool for Delayed Cerebral IsChemia	Co-Investigator	3,064,122	300,000	2023–2028

Completed

	Funder	Title	Role	Total Budget	My Budget	Dates
G9	American Heart Association (Career Development Award) 24CDA1274465	Deep Learning Framework for Non-Invasive ICP Waveforms	Principal Investigator (Waived upon Faculty Independence award)	231,000	231,000	2024–2027
G8	National Institutes of Health (NIH R21 NS113055)	Machine Learning for Acute Hydrocephalus	Co-Investigator	445,500	100,000	2020–2022
G7	American Heart Association Postdoctoral Fellowship (20POST35210653)	Real-Time Detection of Delayed Cerebral Ischemia	Principal Investigator	135,352.00	135,352.00	2020-2021
G6	National Institutes of Health (NIH K01-ES026833)	Multi-parametric Prediction of Vasospasm after SAH	Key Personnel	1,08,000	200,000	2016–2020
G5	NVIDIA	Prediction of DCI using Deep Learning	Principal Investigator	GPU	-	-

TEACHING

Courses Taught

Instructor: COMSB 1016: Introduction to Data Science and Computation Thinking Barnard College, Columbia University, Department of Computer Science	Fall 2026
Instructor: COMSB 3997: Artificial Intelligence in Medicine Barnard College, Columbia University, Department of Computer Science	Spring 2026
Instructor: COMSB 1016: Introduction to Data Science and Computation Thinking Barnard College, Columbia University, Department of Computer Science	Fall 2025
Guest Lecture: Lecture Series, EEG data analysis Columbia University Medical Center, Department of Neurology	Fall 2018
Guest Lecture: Deep Learning Series II Columbia University Medical Center, Department of Biomedical Informatics	Fall 2018

Guest Lecture: Deep Learning Series I
Columbia University Medical Center, Department of Biomedical Informatics

Spring 2018

Teaching Assistant: ECE 3331: Programming in C++
University of Houston, Department of Electrical and Computer Engineering

Spring 2014

Teaching Assistant: ECE 6342: Digital Signal Processing
University of Houston, Department of Electrical and Computer Engineering

Fall 2013

Student Advising

Undergraduate, Academic Advising

Barnard College, Department of Computer Science.

Student	Key Dates
Sofia Brauweiler	2025–
Valeria Olguin	2025–
Lilia Shmulenson	2025–
Julia Wang	2025–
Katie Wang	2025–
Sherry Wang	2025–
Anissa Basu	2026–
Xiaoyan Ding	2026–
Rafia Hossain	2026–
Youqiao Sun	2026–

Research Advisor

Student	Department	Notes	Key Dates
Aayushi Ranjan	Computer Science	Winner Summer Research grant at Barnard College. Working on analyzing neural correlates of music on n back task	2025–
Pari Bhandari	Computer Science	Winner Summer Research grant at Barnard College. Working on developing multi-modal data analysis models for predicting sepsis using MIMIC IV datasets.	2026–

Other Significant Mentoring (PhD, Postdoctoral, Post-Undergraduate)

Student	Department	Notes	Key Dates	Current Position
Kalija Terilli, B.S.	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2016–2019	Medical Student, Stanford University School of Medicine (Neurology Residency, Mt. Sinai West, 2026)
John Carlson	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2019–2020	Medical Student, Rutgers University

Dan Nametz	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2022–2025	Data Scientist, NYU Langone Health
Jiayu Yao, Ph.D.	Neurology, Columbia University Medical Center	Postdoctoral co- mentor	2024–2026	Data Scientist, Capital One
Tammam Alalqum	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications; selected for M.S. at DBMI, Columbia	2024–	M.S. Student, Dept. of Biomedical Informatics, Columbia University
Zoe Zhou	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2024–	Data Scientist, Dept. of Neurology, Columbia University
Justin Zheng	Computer Science, Columbia University	M.S. research advisor	2025–	M.S. in Data Science, Columbia University
Gisselle Garsi	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2025	Post-bac Premed Program
Yanwei Li	Neurology, Columbia University Medical Center	Research advisor; co- author on multiple journal publications	2025–	PhD Student

Invited Talks

- Innovative Models for Physiological data Analysis (Using FM-WAVES: Foundation Models for Waveform Analysis and Visualization for Enhanced Signal processing), Neurocritical Care Conference, 2025, Montreal, Canada
- Addressing Bias in AI/ML – Ensuring Fairness and Equity in Neurocritical Care, Neurocritical Care Conference, 2024, San Diego, CA
- Foundation models for analyzing high frequency time series data, Barnard College, Department of Computer Science, NY
- I-WAVES: Intelligent Waveform Analysis and Visualization for Enhanced Signals in Neurocritical Care, Indian Institute of Technology, Koita Center of Digital Health, Indian Institute of Technology, Bombay, 2025
- Data-Driven Methods for Analysis of Biological datasets, School of Public Health, University of Texas, 2018

SERVICE & LEADERSHIP

Departmental Service

AI Minor Committee Member, Barnard College, 2026 - Contributing to development and approval process for new AI minor.

Review (Journals and Conferences)

- Review Editor, Editorial Board of Neurocritical and Neurohospitalist Care, *Frontiers in Neurology* (2023–present)
- Critical Care Medicine
- Stroke
- Machine Learning for Healthcare (MLHC) (2016–present)
- Neurocritical Care (2017–present)
- IEEE Journal of Biomedical Health Informatics
- Neuroinformatics
- IEEE Geoscience & Remote Sensing Letters
- Entropy
- PLoS ONE
- eLife

Review (Grants)

- American Heart Association 2019- Present

SCHOLARSHIP: PEER-REVIEWED PUBLICATIONS

48 peer-reviewed publications (40 journal articles, 6 conference papers, 2 in progress). 16 first-author papers. Venues include *NEJM AI*, *Annals of Neurology*, *Stroke*, *Critical Care Medicine*, *NeurIPS*.

Publications in Progress

1. Megjhani M, Morrill, T, Zemel R, Park, S. Context-Aware Mixture of Experts (CAMOE) for predicting delayed cerebral ischemia after Subarachnoid Hemorrhage
2. Z. Zhou, Megjhani, M, Park, S. Predicting Shunt Dependency Using Early Cerebrospinal Drainage and Ventricular Volume

Journal Articles (* denotes equal contribution)

3. Megjhani M, Li Y, Grassi G, et al. ICP-WAVES: Intracranial Pressure Waveform Analysis and Visualization for Enhanced Signal Processing. *IEEE Transactions on Biomedical Engineering* (2026)
4. Park S, Li Y, Megjhani M. Physiological Signals Boosted by Artificial Intelligence Save Time and Brain Function. *NEJM AI*, 2(12):Alra2500585 (2025)
5. Megjhani M, Weinerman B, Alalqum T, et al. Application of a time series foundation model to noninvasively estimate intracranial pressure. *Journal of Clinical Monitoring and Computing*, 39(6):1283–1292 (2025)
6. Balch JA, Brandberg JG, Andris RT, Mesner IG, Nametz DC, Megjhani M, et al. Equipping mathematical models for hospital dynamics using information theory. *npj Digital Medicine*, 8(1):652 (2025)
7. Ku Y, Megjhani M, Alalqum T, et al. Pressure Reactivity and Cerebral Oximetry Indexes in Aneurysmal SAH: A Pilot Study. *Neurocritical Care* (2025)
8. McLaren SH, Camargo CA, Florin TA, Leu CS, Mejias A, Han J, Megjhani M, et al. Young infants with bronchiolitis at low risk of respiratory deterioration: prospective cohort study protocol. *BMJ Open*, 15(8):e101880 (2025)
9. Kwon SB, Weinerman B, Nametz D, et al. Pulse rate variability as a predictor for LOS in bronchiolitis PICU patients. *Journal of Clinical Monitoring and Computing*, 39(3):533–539 (2025)
10. Weinerman B, Kwon SB, Alalqum T, et al. Pediatric VA-ECMO parameters to identify early risk factors for mortality. *Perfusion* (2025)
11. Kwon S, Weinerman B, Nametz D, Megjhani M, et al. Non-invasive pulse arrival time and cardiac index in pediatric heart transplant. *Physiological Measurement*, 45(7):07NT01 (2024)

12. Ridha M, Megjhani M, Nametz D, et al. Suboptimal Cerebral Perfusion Associated with Ischemia After ICH. *Neurocritical Care* (2023)
13. Kwon SB, Megjhani M, Nametz D, et al. Heart rate and HRV as prognostic features after cardiac arrest: scoping review. *Resuscitation Plus*, 15:100450 (2023)
14. Megjhani M, Terilli K, Kwon SB, et al. Automatic identification of ICP waveform during EVD clamping via wavelet analysis. *Physiological Measurement*, 44(6):064002 (2023)
15. Megjhani M, Terilli K, Weinerman B, et al. A Deep Learning Framework for Noninvasive ICP Waveforms from TCD. *Annals of Neurology*, 94(1):196–202 (2023)
16. Kastenholz N, Megjhani M, Conzen-Dilger C, et al. Oxygen Reactivity Index and Disturbed Perfusion Regulation after SAH. *Critical Care*, 27(1):235 (2023)
17. Ranard BL, Vodovotz Y, Asgari S, Megjhani M, et al. Mathematical model of SARS-CoV-2 immunity predicts Paxlovid rebound. *Journal of Critical Care*, 74:154210 (2023)
18. Gomez LA, Shen Q, Doyle K, et al. Classification of Level of Consciousness in a Neurological ICU Using Physiological Data. *Neurocritical Care*, 38(1):118–128 (2023)
19. Megjhani M, Weiss M, Ford J, et al. Optimal CPP and brain tissue oxygen in aneurysmal SAH. *Stroke*, 54(1):189–197 (2023)
20. Ranard B, Megjhani M, Terilli K, et al. Heart rate variability and adrenal size as clues to sudden cardiac death in COVID-19. *Journal of Critical Care* (2022)
21. Weiss M, Albanna W, Conzen-Dilger C, Kastenholz N, et al. Intraarterial nimodipine versus induced hypertension for DCI. *Stroke*, 53(8):2607–2616 (2022)
22. Rubinos C, Kwon SB, Megjhani M, et al. Predicting Shunt Dependency from CSF Drainage Effect on Ventricular Size. *Neurocritical Care* (2022)
23. Megjhani M, Weiss M, Kwon SB, et al. Vector Angle Analysis for Continuous Prediction of DCI. *Neurocritical Care* (2022)
24. Weiss M, Albanna W, Conzen C, Megjhani M, et al. Optimal CPP During DCI after Aneurysmal SAH. *Critical Care Medicine*, 50(2):183–191 (2022)
25. Megjhani M, Terilli K, Kalaspudi L, et al. Dynamic ICP Waveform Morphology Predicts Ventriculitis. *Neurocritical Care* (2021)
26. Ranard B, Megjhani M*, Terilli K*, et al. Identification of Endotypes of Hospitalized COVID-19 Patients. *Frontiers in Medicine*, 8 (2021)
27. Megjhani M, Terilli K*, Weiss M, et al. Dynamic Detection of DCI: A Study in Three Centers. *Stroke*, 52(4):1370–1379 (2021)
28. Megjhani M, Terilli K, Kaplan A, et al. Clustering to Investigate Changes in ICP Waveform Morphology in Ventriculitis. *Physiological Measurement*, 40(1):015002 (2019)
29. Rasulo F, Piva S, Park S, Oddo M, Megjhani M, et al. Peri-hemorrhagic Metabolites and Cerebral Hemodynamics in ICH. *Frontiers in Neurology* (2020)
30. Savarraj J, Hergenroeder G, Zhu L, Chang T, Park S, Megjhani M, et al. ML to predict DCI and outcomes in SAH. *Neurology* (2020)
31. Alkhachroum A*, Terilli K*, Megjhani M*, Park S. Harnessing Big Data in Neurocritical Care in Precision Medicine. *Current Treatment Options in Neurology*, 22(5) (2020)
32. Stember JN, Terilli KL, Perez E, Megjhani M, et al. Surface Point Cloud Ultrasound with TCD Coregistration. *Journal of Digital Imaging* (2020)
33. Alkhachroum A, Megjhani M, Terilli K, et al. Hyperemia in SAH Associated with Increased Risk of Seizures. *Journal of Cerebral Blood Flow & Metabolism* (2019)
34. Claassen J, Doyle K, Matory A, et al. Detection of Brain Activation in Unresponsive Patients with Acute Brain Injury. *New England Journal of Medicine*, 380:2497–2505 (2019)
35. Megjhani M, Kaffashi F, Terilli K, et al. Heart Rate Variability as a Biomarker of Neurocardiogenic Injury after SAH. *Neurocritical Care* (2019)
36. Rohaut B, Doyle K, Reynolds A, Megjhani M, et al. Deep Structural Brain Lesions and Consciousness Impairment in Hemorrhagic Stroke. *Scientific Reports*, 9(1):4174 (2019)
37. Megjhani M, Terilli K, Park S, et al. Active Learning Framework for ICP Waveform Identification. *Physiological Measurement* (2018)
38. Park S, Megjhani M, Claassen J, Elhadad N. Predicting DCI after SAH Using High-Frequency Physiological Data. *Journal of Clinical and Medical Computing* (2018)

39. Megjhani M, Terilli K, Claassen J, Elhadad N, Park S. Computational Dictionary Learning Improves DCI Prediction. *Frontiers in Neurology* (2018)
40. Megjhani M, Terilli K, Martin A, et al. Deriving PRx and CPPopt from 0.2-Hz Data. *Acta Neurochirurgica Supplement, vol 126, Springer* (2018)
41. Cruz-Garza J, Brantley J, Nakagome S, Megjhani M, et al. Mobile EEG in Art Museum Setting: Signal Quality and Usability. *Frontiers in Human Neuroscience* (2017)
42. Megjhani M, Roysam B. Morphological Constraint Spectral Unmixing of Biological Tissues using Confocal Microscopy. *Bioinformatics* (2017)
43. Barton E, Lu Y, Megjhani M, et al. Binge Alcohol Alters Exercise-Driven Neuroplasticity. *Neuroscience*, 343 (2017)
44. Megjhani M*, Kontson K*, Cruz-Garza JG, et al. Your Brain on Art: Emergent Cortical Dynamics During Aesthetic Experiences. *Frontiers in Human Neuroscience* (2015)
45. Megjhani M, Rey-Villamizar N, Merouane A, et al. Population-scale 3D Reconstruction of Microglia Arbors. *Bioinformatics*, 31(14):2190–2198 (2015)
46. Xu Y, Megjhani M, Shain W, et al. Unsupervised Profiling of Microglial Arbor Morphologies. *IEEE Journal of Selected Topics in Signal Processing* (2015)
47. Rey N, Somasundar V, Megjhani M, et al. Large-scale Automated Image Analysis of Brain Tissue Surrounding Implanted Devices. *Frontiers in Neuroinformatics*, 8(39) (2014)

(Selected) Peer-Reviewed Conference Papers & Abstracts

48. Morrill T, Puli A, Megjhani M, Park S, Zemel R. Let the Experts Speak: Improving Survival Prediction via Mixture-of-Experts Heads. *NeurIPS* (2025)
49. Elhussein A, Megjhani M, Kwon SB, et al. Generalizable Physiological Model for DCI Detection using Federated Learning. *IEEE BIBM*, pp. 1886–1889 (2023)
50. Cao H, Finer J, Megjhani M, et al. ML Model Deployment Using Real-Time Physiological Monitoring for DCI Detection. *IEEE HI-POCT*, pp. 42–45 (2022)
51. Magid-Bernstein J, Megjhani M, et al. CSF Cell Index in Predicting Shunt Dependence after ICH. *Neurocritical Care Society* (2020)
52. Megjhani M, Terilli K, Kaplan A, et al. ICP Waveform Morphology Changes in Ventriculitis. *ICP Symposium* (2019)
53. Ford J, Megjhani M, et al. Brain Tissue Oxygenation for Optimal CPP in SAH. *American Academy of Neurology* (2018)

SOFTWARE & TOOLS

ICP-WAVES: Deep learning framework for non-invasive intracranial pressure waveform estimation from TCD and ABP signals. Multi-center validated. (Megjhani et al., *Annals of Neurology* 2023; *IEEE TBME* 2026)

FM-WAVES: Foundation model for physiological waveform analysis, pre-trained on heterogeneous ICU time series for zero-shot and few-shot clinical prediction tasks.

DCI Detection Pipeline: Real-time multivariate physiological monitoring system for delayed cerebral ischemia, deployed in prospective three-center study (*Stroke* 2021).

AWARDS & HONORS

Future Faculty Fellowship; University of Houston, TX	2015
Tuition Fellowship for PhD Program; University of Houston, TX	2012–2016
Distinguished Student Award (State Rank), Osmania University, India	2005

ACADEMIC & PROFESSIONAL POSITIONS

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- Roman Family Faculty, Dept. of Computer Science

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| Barnard College, Columbia University | Sep 2025 – Present |
| • Associate Research Scientist (equiv. Asst. Professor),
Dept. of Neurology Columbia University Irving Medical Center | Feb 2021 – Present |
| • Postdoctoral Research Scientist,
Dept. of Neurology, Columbia University Irving Medical Center | Aug 2016 – Feb 2021 |
| • Research Scientist,
Brain-Machine Interface Lab, University of Houston, TX | Aug 2014 – Aug 2016 |
| • Research Scientist,
Bio-Image Analytics Lab, University of Houston, TX | Jan 2012 – Aug 2016 |

Non-Academic Positions

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| • Senior Associate, Cognizant Technology Solutions, San Jose, CA | Mar 2009 – Jan 2012 |
| • Associate, Cognizant Technology Solutions, San Jose, CA | Nov 2005 – Mar 2009 |